Methods of cleaning of sewers

a. Cleaning and flushing

b. Cleaning of catch pits, etc.

Activity 1

Collect pictures of various methods of cleaning of sewers and prepare an album

5.4.7 Sewer appurtenances

They are the additional structures constructed at suitable intervals along the sewer line. They are constructed for the efficient working and maintenance of sewerage system.

The important sewer appurtenances are as follows

1. Manhole
2. Drop Manhole
3. Lamp hole
4. Catch basins
5. Clean out
6. Flushing tanks
7. Inlets
8. Grease and oil traps
9. Inverted syphons
10. Storm water regulators

5.5 Quality of Sewage

The quality of sewage plays an important role in the design of sewer and construction of various treatment units. Also the quality of sewage decides the method of treatment to be adopted.

5.5.1 Properties of Sewage

1. Physical characteristics
2. Chemical characteristics
3. Biological characteristics

1. Physical Characteristics

The sewage has the following physical characteristics

a. Odour

Fresh sewage is odourless. When it becomes stale, it has an offensive odour by hydrogen sulphide and other sulphur compounds.

b. Colour

If the sewage is fresh the colour of the sewage is yellowish grey or light brown. If the sewage is septic or stale, then the colour is black or dark.

c. Turbidity

The turbidity of the sewage directly depends on the quantity of solid matter present in suspension. Sewage is normally turbid.

d. Temperature

The normal temperature of the sewage is slightly higher than the water supplied for various uses. When the sewage flows through closed conduits, its temperature further rises.

This affects the biological activity, solubility of gases and viscosity of sewage.

2. Chemical Characteristics

This indicate the state of sewage decomposition, its strength and types of treatment required. Fresh sewage is alkaline and good for bacterial action. Stale or septic sewage is acidic and difficult to treat efficiently.

The chemical characteristics depends upon the substances contained in the sewage. The substances are complex organic (derived from urine, human-excreta) and inorganic chemicals.

3 Biological Characteristics

It is necessary to know the biological characteristics of sewage due to the

U5.indd 94 12/11/2019 1:03:05 PM